

KnowledgeGraph.AI

Intelligent Educational Graph Synthesizer & RAG Companion

1. Product Overview

KnowledgeGraph.AI is a state-of-the-art educational tech platform designed to transform unstructured textbooks, lecture notes, and learning documents (PDFs) into interactive, visual learning pathways. By combining natural language processing, vector search indexers, and graph theory algorithms, the application generates a personalized visual conceptual model of any academic document. Students can navigate concepts interactively, run pathway animations, engage in question-answering with an AI tutor grounded in the text, and generate structured study roadmaps.

2. Processing Pipeline Architecture

When a user uploads a PDF document through the dashboard, it triggers a background process orchestrated through an advanced asynchronous queue structure. The processing follows six key stages:

Stage A: Text Extraction

The server processes the uploaded PDF binary using a page-by-page rendering pipeline, extracting the raw textual character stream and compiling metadata such as total page counts.

Stage B: Page-by-Page Chunking

The raw text is segmented into distinct semantic chunks mapped to specific pages, ensuring that context window sizes are optimized for downstream AI processing and citation reference lookup.

Stage C: AI Concept Extraction

The text chunks are analyzed using Google Gemini 1.5 Flash (or local rules-based mock modes when offline). The AI extracts key educational concepts, difficulty scores (1-10), importance levels, and concise summaries, merging duplicates dynamically using a concept normalizer database.

Stage D: AI Relationship Discovery

Concepts are correlated to discover structural connections. The system extracts relationships such as 'prerequisite' (e.g., Algebra -> Calculus) or 'related_to'. Edge links are verified against document concepts and filtered using confidence scores (threshold ≥ 0.70).

Stage E: Topology Graph Construction

The backend compiles concepts (nodes) and relationships (edges) into a network layout snapshot. The coordinates are mathematically calculated to layout nodes from left (lower difficulty) to right (higher difficulty) and group them vertically by academic categories.

Stage F: Vector Index Generation

High-dimensional word embeddings are calculated for all chunks and concepts using text-embedding models. These vectors are indexed in MongoDB to support real-time semantic search and RAG Q&A.

3. Interactive Feature Modules

A. Dashboard & Document Hub

The homepage of logged-in users lists all processed textbooks and learning materials. Users can upload new files (up to 15MB) and monitor extraction progress in real-time via progress bars and stage labels (e.g., 'Assembling Topology Graph'). Once completed, shortcuts open the Graph, Chat, or Roadmaps pages.

B. Graph Viewer (React Flow Canvas)

The core visual tool renders the topological knowledge map of the document. Key details include:

1. Interactive Nodes: Concept boxes color-coded by academic categories (e.g., Pink for Deep Learning).
2. Premium Styled Edges: Smoothstep lines showing prerequisites (dashed red) and general links (blue).
3. Edge Labels: Floating 'prerequisite' indicators styled with dark theme slate backgrounds and edge-matching borders.
4. Concept Inspector Sidebar: Clicking any node pulls up its metadata, occurrence frequency, and highlights the precise source text paragraphs with clickable page citations.
5. Exact, Fuzzy, and Semantic Search: Highlights matching concepts instantly with glowing indigo outlines.
6. Visual Traversal DSA: Students select a starting topic, choose BFS (Breadth-First Search) or DFS (Depth-First Search), and click 'Animate Path' to watch the logical learning sequence light up step-by-step.

C. RAG Chat Assistant

A conversational tutoring chatbot grounded in the document contents. It pulls relevant text chunks using semantic similarity search on the user question, synthesizing a clear educational response with highlighted concepts and inline text citations.

D. Study Roadmaps (Learning Path Generator)

Generates step-by-step custom curriculum plans for any selected target topic. The algorithm calculates prerequisite dependencies recursively and outputs sequential study milestones with estimated study times, category descriptions, and prerequisites.